

AgriConnect (SIH PS 26132)

Platform Features, Technical Capabilities & System Architecture Report

Problem Statement: Strengthening market linkages & price discovery for farmers (Govt of Maharashtra | SIH 26132)

Core Tech Stack: FastAPI (Python) + React (Vite) + SQLite/PostgreSQL + Bhashini Voice AI + MobileNet Vision

Target Audience: Indian Farmers, FPOs, Grain Buyers, State Marketing Boards, NABARD Officers

1. Executive Summary & Core Platform Foundations

AgriConnect is a neutral digital agriculture layer built directly on top of India's public **Agmarknet (data.gov.in)** and **e-NAM** daily commodity market price datasets. It solves the critical problem of asymmetric price discovery and distress sales by providing explainable AI advisories, computer-vision quality checks, direct buyer matchmaking, and SHA-256 tamper-proof ledger receipts.

2. Comprehensive List of Features & Functionalities

■ Multi-Role & Personas

- **3 User Roles:** Farmer / FPO Representative (Ramesh Kumar), Verified Buyer (Rajesh Agro Traders), Admin Moderator (Agmarknet Admin).
- **Multi-Auth Engine:** Standard Google OAuth 2.0 SSO button, 6-digit Mobile SMS OTP (demo code 123456), and Email/Password.
- **Dual-Mode UI Toggle:** Top navbar pill toggle switching between ■■■ *Kisan View* (simplified Bharat UX) and ■ *Technical / Judge View* (exposes raw formulas & cryptographic hashes).

■ Price Discovery & AI Advisory Engine

- **Live Mandi Dashboard:** Real-time Agmarknet rates for 12+ crops (Wheat, Paddy, Onion, Potato, Tomato, Chana, Soyabean, Cotton, etc.) across 8+ major mandis (Azadpur, Nashik, Kanpur, Indore, Agra, etc.).
- **Explainable Advisory Engine:** Rule-based engine comparing current prices with 7-day and 30-day moving averages and arrival volumes, outputting plain-language reasoning (*SELL NOW* vs *HOLD 3 DAYS*).
- **AI Time-Series Price Forecast:** 7-14 day price projection model with **85% confidence upper/lower bounds** rendered on Recharts with seasonal market insights.

■ Digital Lot Creation & AI Quality Check

- **3-Step Visual Listing Wizard:** Tactile crop photo tiles, quantity steppers (quintals), price inputs, and mandi location selectors.
- **MobileNet Crop Quality Grading:** Computer vision model calculating indicative **Grade A / B / C** ratings with defect metrics (size uniformity %, moisture content %, surface spots).

■ AI Buyer Matchmaking & Trade Pipeline

- **Top 5 Buyer Matchmaker:** Ranks verified buyers using a 4-factor scoring model: 35% Crop Match + 25% OpenStreetMap Haversine Distance + 20% Quantity Fit + 20% Buyer Rating.
- **Negotiation Loop:** Binding buyer offers, farmer counter-proposals, and pipeline tracking (*Listed* → *Offered* → *Accepted* → *Paid/Delivered*).
- **UPI Payment Integration:** Settlement tracking capturing official UPI transaction reference numbers (*UPI/329481048201/PAYMENT_DONE*).

■ SHA-256 Immutable Hash Ledger

- **Append-Only Cryptographic Chain:** Sequential block hashing where $Current\ Hash = SHA256(Previous\ Hash + Payload)$.
- **Green Digital Rasid:** Printable cryptographic sale certificate with QR code and collapsible SHA-256 hash accordions.
- **Ledger Auditor Tool:** Online integrity checker verifying zero-tampering across all recorded agricultural trades.

■■ Bhashini Voice AI & Regional Languages

- **Multi-Language Voice Chat:** Real-time Speech-to-Text (STT) and Text-to-Speech (TTS) supporting (Hindi), English (India), and (Marathi).
- **Direct 1-Click Voice Chips:** Preset query buttons (*Kanpur Wheat Rate*, *Sell 50 Qtl Onion*, *Weather Alert*) that auto-speak answers out loud.
- **Persistent Voice FAB:** Pinned bottom-right floating microphone button (*Bol Kar Bhaav Jaanein*).

■ Weather, Schemes, SMS & Admin Monitoring

- **IMD Weather Alerts:** Heavy rain and pest risk warnings (purple blotch / late blight) with agronomic preventive advice.
- **Govt Scheme Recommender:** Profile matcher for PM-KISAN, PMFBY, KCC, SMAM, PM-KUSUM linking directly to myScheme.gov.in.
- **56161 SMS Simulator:** Offline feature phone shortcode simulator supporting SMS text commands (*PRICE WHEAT KANPUR*).
- **NABARD & State Board Dashboard:** Macro analytics for total trade volume, average realized farmer price, price uplift %, and fraud anomaly alerts.

3. Technical Architecture Summary Matrix

Module	Tech / Algorithm	Inputs	Output / Value
Price Data	Agmarknet daily API + CSV snapshot	Commodity, Mandi, Date	Modal, Min, Max rate & Arrival Qtl
Price Forecast	Time-Series Statistical Model	60-day price/arrival history	7-14 day curve + 85% confidence band
Quality Grade	MobileNet Classifier	Crop produce photos	Grade A/B/C + Defect Metrics
Buyer Match	Haversine + Weighted Scoring	Crop, Qty, GPS coords	Top 5 ranked verified buyers (% fit)
Voice AI	Bhashini STT + Web Speech TTS	Spoken Hindi/English audio	Auto-spoken natural language response
Hash Ledger	SHA-256 Cryptographic Chain	Trade record + prev hash	Immutable Hash + Sale Certificate

4. End-to-End System Flowchart & Journey

The diagram below details the operational journey of a transaction through AgriConnect, from authentication to cryptographic settlement:



